3. Gas Laws

The equation that says how hard it pushes

Heat a trapped gas and the pressure rises in proportion

Lesson 3 · about 75 minutes · everyone — kitchen care only

What you need

- A balloon, a bottle, a bowl of hot water and one of iced water
- A syringe with the nozzle blocked, and weights
- A pressure gauge and a bicycle pump
- A thermometer
- A marshmallow and a syringe, for entertainment

Predict first

A sealed rigid container of air at room temperature is heated from 20 °C to 1000 °C, which is roughly what a burning fuel-air mixture reaches. The volume cannot change. By what factor does the pressure rise? Guess before you calculate — then calculate.

Do it

1. **Balloon on a bottle.** Stretch a balloon over an empty bottle's neck. Stand the bottle in hot water — the balloon inflates. Then iced water — it is drawn in. Nothing entered or left.
2. **Squeeze a syringe.** Block the nozzle, push the plunger, and feel the resistance rise. Halve the volume and the force roughly doubles. That is Boyle's law under your thumb.
3. **Marshmallow.** A marshmallow in a blocked syringe. Pull the plunger and it swells; push and it shrinks. The gas trapped in its foam is obeying the same rule.
4. **Measure it properly.** Pump a fixed volume to a known pressure, note the gauge, and warm the vessel. Record pressure against temperature in kelvin and plot it. It is a straight line, and extrapolating it backwards points at absolute zero.
5. **Do the launcher arithmetic.** Your chamber is fixed volume. Take the flame temperature of a hydrocarbon in air as roughly 2000 K and room temperature as 293 K, and work out the pressure ratio.

What it means

The gas law	$PV = nRT$. Pressure times volume equals amount times temperature, with a constant. Fix the volume and the amount, and pressure is simply proportional to *absolute* temperature.
Kelvin, not Celsius	The proportionality only works from absolute zero. Going from 20 °C to 40 °C is not a doubling — it is 293 K to 313 K, about 7%. This is the mistake everyone makes once.
The launcher number	From 293 K to roughly 2000 K is a factor of about seven in absolute temperature. In practice a combustion launcher chamber sees a peak of roughly 3 – 6 atmospheres absolute, because the burn is not instantaneous, the potato starts moving, and heat is lost to the walls almost immediately.
And n changes too	Burning also changes the number of gas molecules. That is a second, smaller contribution on top of the temperature effect.
Why that is survivable	Schedule 40 pressure PVC is rated for a sustained working pressure well above that, and the launcher's peak lasts a few thousandths of a second. The margin is real — but it is a margin, not a licence. It is also why over-fuelling, cold pipe, and un-cured joints all matter. *see the **Safety** sheet*
Force, not pressure, moves the potato	Pressure times area. Three extra atmospheres on a 1.5 in bore is roughly 80 lb pushing on the potato's base, for the whole length of the barrel.

Break it

Longer barrel	The same force acting over a longer distance does more work, so the potato leaves faster — up to the point where the gas has expanded so much that pressure has fallen and friction wins. Build a second barrel and find out where that is.
Bigger bore	More area, so more force — but also more gas volume needed to fill it, so the pressure falls faster as the potato moves. Argue both sides before you assume bigger is better.
Cold day	Starting at 0 °C instead of 20 °C is only a 7% difference in the gas law — but it makes PVC substantially more brittle, which matters far more. *see the* ***Safety*** *sheet*
Work it out first	Energy delivered is roughly average force times barrel length. Compare that with $\frac{1}{2}mv^2$ for a 200 g potato and predict the muzzle velocity you will measure in Lesson 4.

The link to the launcher

This lesson is what makes the project engineering rather than a stunt. Before you fire it you can put a number on the pressure, a number on the force, and a number on the expected muzzle velocity — and then in Lesson 4 you measure the velocity and see how close you were.